Experiment 6

Creation and Deployment of Smart Contract and Transactions using Solidity

Aim

To understand and deploy the `Ballot` voting smart contract (the Remix IDE sample), changing the proposal names from `bytes32` to `string`, and to exercise its full flow of transactions: giving the right to vote, delegating, voting, and declaring a winner.

Theory

`Ballot.sol` implements electronic voting with delegation. The chairperson (the deploying account) creates one `Proposal` per name and gives individual addresses the right to vote; each voter has a `weight` (1 by default, increased when others delegate to them), a `voted` flag, an optional `delegate`, and the index of the proposal they voted for. `vote()` adds the voter's weight to a proposal's `voteCount`; `winningProposal()` returns the index with the highest count, and `winnerName()` returns its name.

Change made for this experiment: the original sample stores each proposal name as `bytes32`, so deploying it in Remix needs every name converted to a hex string first (hence the web3-type-converter link in the manual). Here `Proposal.name` and the constructor argument are plain `string`, so names such as `"Proposal A: Library Upgrade"` are passed in as-is.

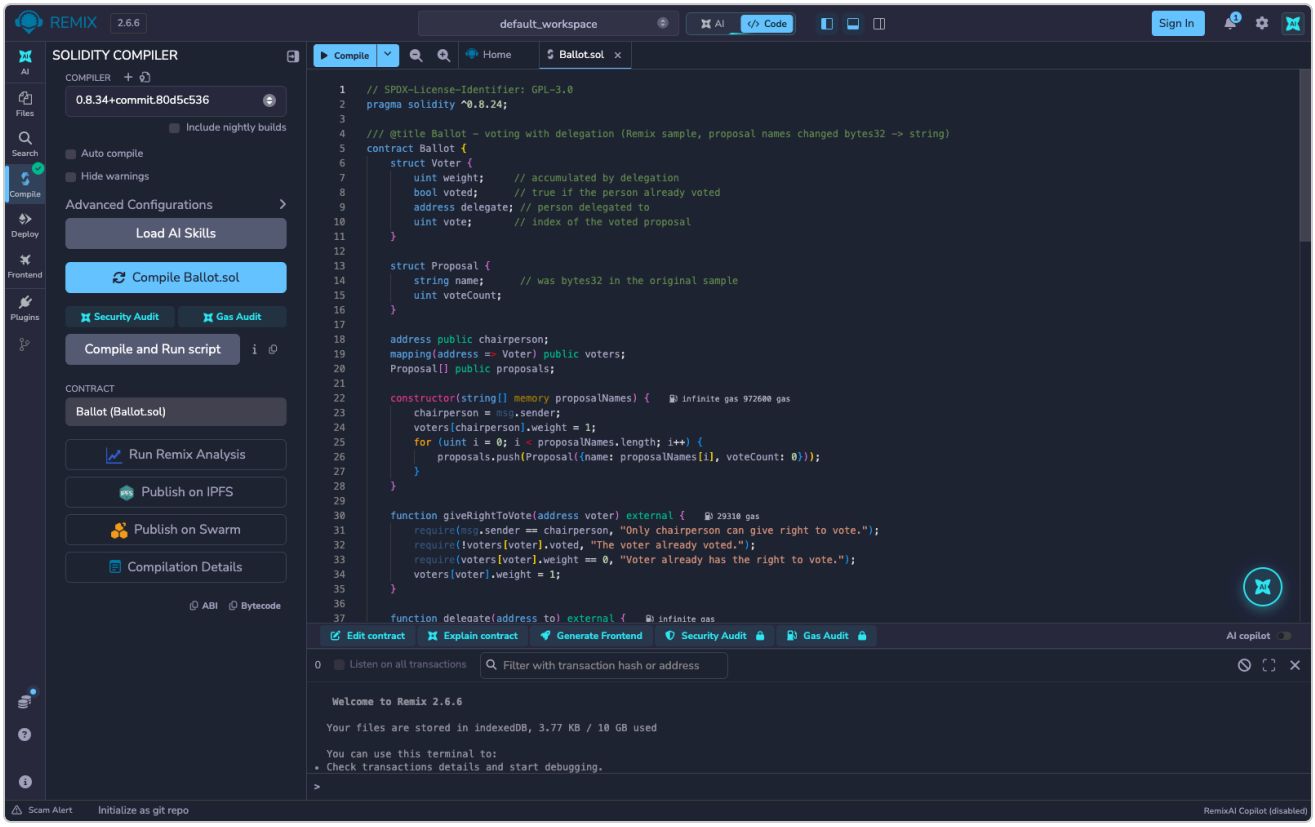
Viva topics

1. **Relevance of `require` statements.** `require(condition, "message")` is a guard: if the condition is false, execution stops, every state change made in that transaction is rolled back, and the message is returned to the caller. In `Ballot` it enforces who may act (`msg.sender == chairperson`, `weight != 0`) and when (`!voted`), so an invalid call can never corrupt the vote counts.

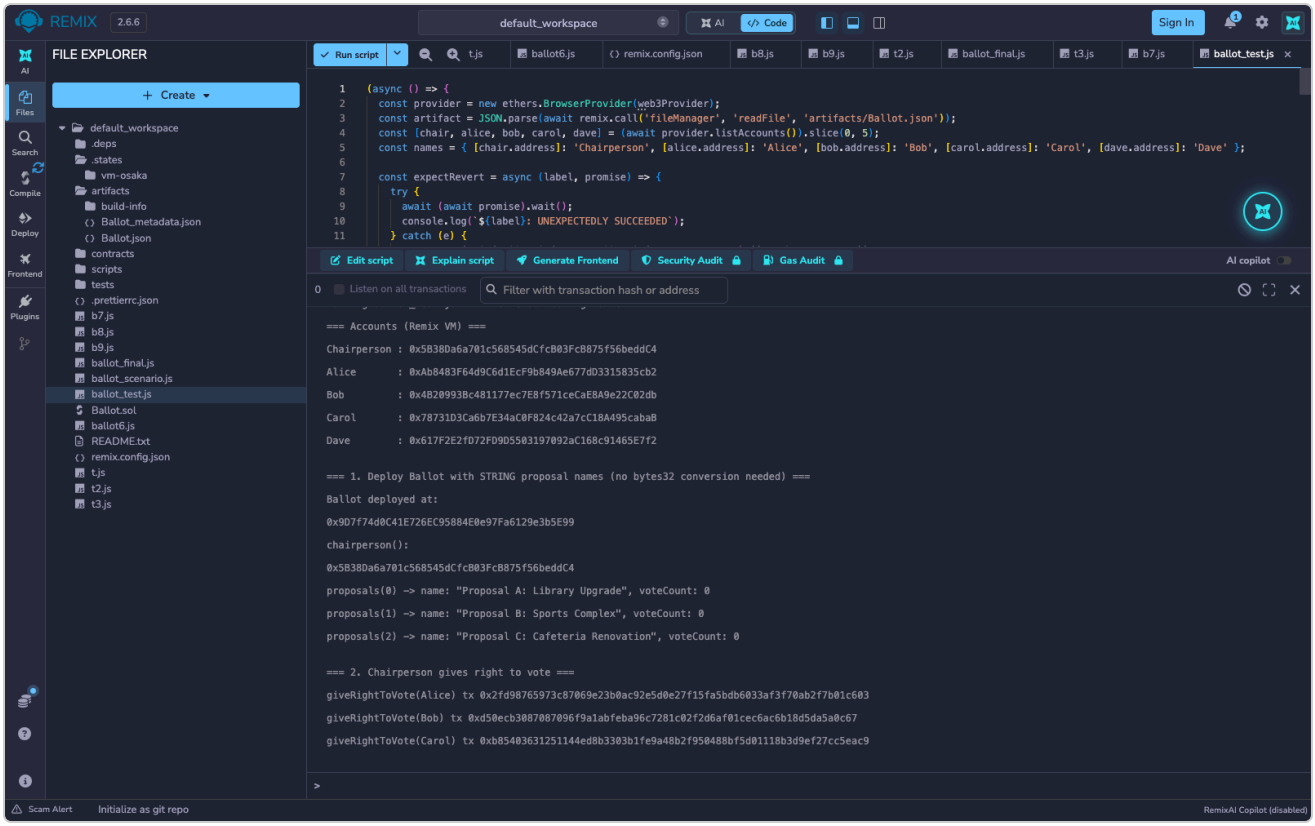
2. **`mapping`, `storage` and `memory`.** A `mapping(address => Voter)` is a hash-table-like key-value store; it has no length and cannot be iterated, but lookups are cheap. `storage` is the contract's permanent on-chain state (writes cost gas); a `storage` local such as `Voter storage sender = voters[msg.sender]` is a *reference* to that state, so edits persist. `memory` is temporary, per-call data that is discarded when the function returns, and is cheaper (e.g. the `string[] memory proposalNames` constructor argument).
3. **Why `bytes32` instead of `string`?** `bytes32` is fixed-size, fits in one storage slot and is cheap to store and compare, but limits names to 32 bytes and needs off-chain conversion for readability. `string` is dynamically sized, human-readable and unlimited in length, but costs more gas and cannot be compared with `==` directly. For a small ballot, readability outweighs the extra gas, which is why this experiment uses `string`.

Tools used: Remix IDE (web, v2.6.6), Solidity compiler 0.8.34 (the contract declares `pragma solidity ^0.8.24`), the in-browser *Remix VM (Osaka)* test blockchain with its pre-funded accounts, and a short JavaScript script run in Remix's own terminal with the built-in ethers.js v6 runner to repeat the whole voting scenario.

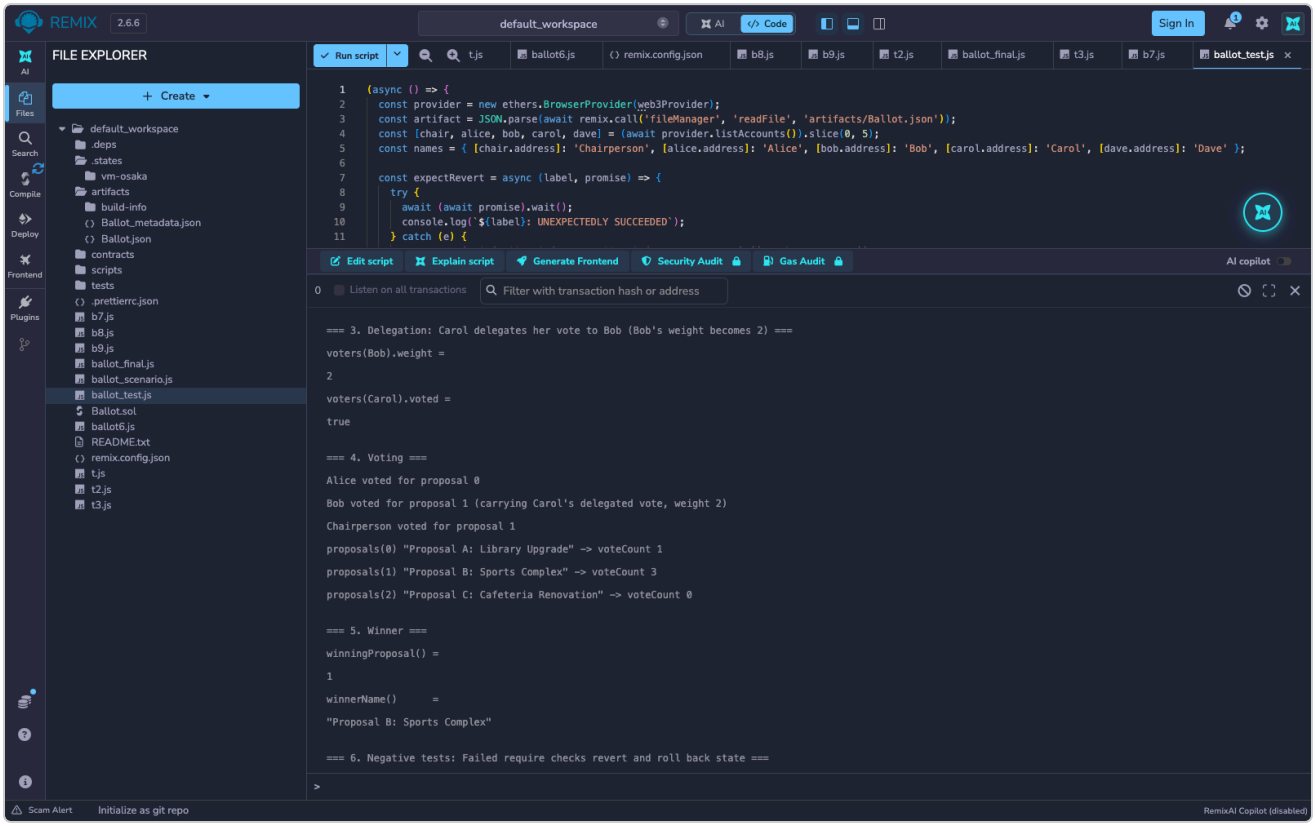
Output



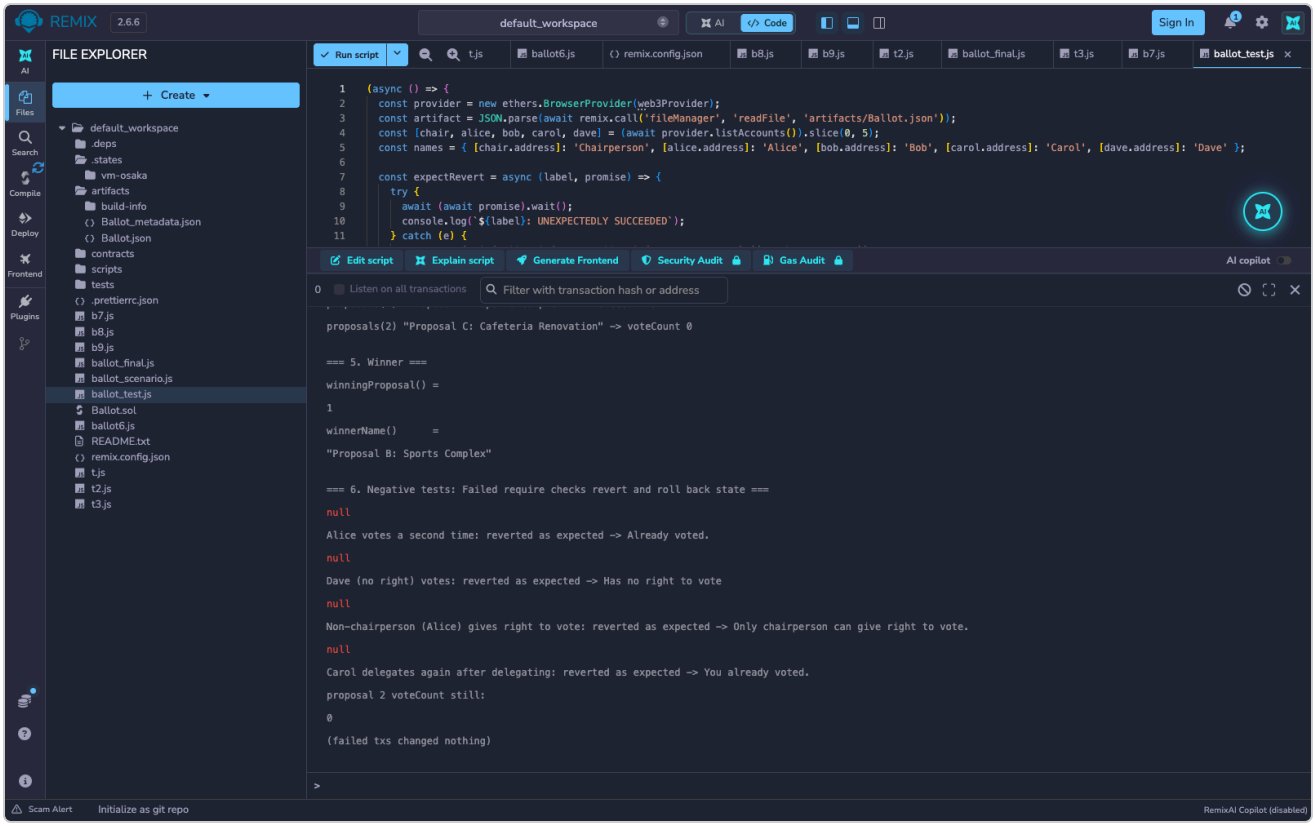
`Ballot.sol` (proposal names changed from `bytes32` to `string`) compiles with no errors in Remix; the green tick on the Solidity Compiler icon confirms it.



The five Remix VM accounts (Chairperson, Alice, Bob, Carol, Dave), then the contract deployed with three readable proposal names passed straight in as strings and read back unchanged from `proposals(i)`, followed by the chairperson giving Alice, Bob and Carol the right to vote.



Carol delegates to Bob, which raises Bob's weight to 2 and marks Carol as having voted. Alice, Bob and the chairperson then vote, giving Proposal B a count of 3 against 1 for Proposal A.



`winnerName()` returns "Proposal B: Sports Complex" as plain text. Four invalid calls (double vote, voting without the right, a non-chairperson granting the right, delegating after already voting) each revert with the contract's own `require` message, and proposal 2's vote count is still 0 afterwards.

Conclusion

This experiment compiled and deployed the Remix `Ballot` contract on the Remix IDE's built-in Remix VM, with the proposal names changed from `bytes32` to `string`. The change removed the need for any off-chain hex conversion: names go in and come out as readable text. A full election was run: three voters were enfranchised, one delegated to another (raising the delegate's weight to 2), votes were cast, and `winnerName()` correctly returned "Proposal B: Sports Complex" with 3 votes against 1. Four negative tests each reverted with the exact `require` message and left the vote counts untouched, confirming how `require`, `mapping`, `storage` references and `memory` arguments combine to keep a voting contract's state correct.